

Pavani Jain

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Education

Northeastern University

Master of Science in Artificial Intelligence | GPA: 3.41/4.00

Expected May 2027

Boston, MA

Guru Gobind Singh Indraprastha University

Bachelor of Computer Applications (BCA) | GPA: 3.64/4.00

Aug 2022 – Jun 2025

Delhi, India

Skills

Languages: Python, R, Java, C, C++, JavaScript, Bash, SQL, HTML, CSS

ML/AI: Deep Learning, NLP, Generative AI, LLMs, RAG, Agentic AI, AI Orchestration, MLOps, Anomaly Detection, Prompt Engineering, A/B Testing

Libraries: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, LangChain, LangGraph, HuggingFace, Transformers, OpenAI API, Anthropic API, SpaCy, PySpark

Data & Tools: PostgreSQL, pgvector, FAISS, MongoDB, BigQuery, Snowflake, ETL, Tableau, Power BI, FastAPI, AWS (SageMaker, S3, EC2, Bedrock), Docker, Kubernetes, Git, MLflow, CI/CD

Experience

AI Engineer Intern

Thingqr

May 2026 – Present

Austin, TX

- Architected the AI orchestration layer (Python, LangChain, LangGraph, OpenAI/Anthropic APIs) powering 5+ multi-agent, multi-turn interview workflows with stateful agent graphs, 8+ tool-calling nodes, and structured output parsing across 3 seniority tiers and 4 role families; built an LLM-as-judge evaluation pipeline scoring 6 competency dimensions over 50+ behavioral signals per session with prompt-engineered guardrails, hallucination checks, and chain-of-thought traces producing reproducible, auditable scorecards.
- Developed a production RAG pipeline (FAISS, pgvector, sentence-transformers, hybrid retrieval) over 100+ scenarios at ~90% top-5 accuracy; integrated with Dockerized Linux workspaces on AWS (EC2, Bedrock) and FastAPI microservices – cutting session spin-up from ~2 min to ~25 sec, P95 LLM latency to ~1.8 sec.

ML Researcher – Northeastern University

Anomaly Detection in Financial Transactions Using Deep Learning

Oct 2025 – Dec 2025

Boston, MA

- Authored a research paper on autoencoder-based fraud detection on imbalanced data (<0.2% fraud), reviewing 25+ works and comparing 4 architectures (LR, RF, XGBoost, Autoencoder) with SMOTE, undersampling, and feature engineering; ran end-to-end experimentation (preprocessing, threshold tuning, ablation studies) and documented metrics (ROC-AUC, precision-recall, F1) with reproducible results per ML research standards.

Data Analyst Intern

Verve Bridge

Jun 2024 – Sep 2024

Delhi, India

- Identified a ~6% payment success-rate gap by writing 50+ SQL queries (CTEs, window functions, joins) on 50K+ transactions, slicing KPIs by payment method, bank, and merchant segment – surfacing 3 underperforming routes that triggered an A/B test on retry logic.
- Built an end-to-end ETL pipeline in Python (Pandas, NumPy, SQLAlchemy) consolidating transaction logs, settlements, and chargebacks from PostgreSQL – cutting manual prep from ~3 hrs to ~20 mins with data-quality checks flagging 10+ weekly exceptions; delivered 3 Tableau dashboards adopted in weekly business reviews covering transaction volume, conversion rates, refund-to-chargeback ratios, and onboarding funnel, enabling self-serve analytics for product, operations, and growth stakeholders.

Projects

Nanochat – GPT Pretraining & LLM Systems Optimization

Python, PyTorch, Transformers

Mar 2026 – Apr 2026

- Built a GPT-style LM from scratch: BPE tokenizer, multi-head attention, training loop, and chat interface; trained on FineWeb-Edu with 8×H100 GPUs, reaching GPT-2-level validation loss in ~3 hrs at ~\$72 compute cost.
- Studied compute-quality tradeoffs by varying transformer depth, learning rate schedules, and batch sizing.

Genomic Text Curation & Topic Grouping

Python, NLP, Scikit-learn, TF-IDF, Topic Modeling

Jan 2026 – Feb 2026

- Built an NLP pipeline extracting genes, variants, and diseases from 50 genomics abstracts; produced 76 knowledge-graph-ready triples (40 variants, 37 genes, 19 diseases) with 84% variant and 78% gene recall.
- Compared TF-IDF + K-Means, NMF, and LDA for topic modeling; identified 5 coherent biomedical clusters.

Activities

Participant, NU Hacks (48-hour hackathon)

Northeastern University

March 2026

Boston, MA

- Built a RAG clinical Q&A prototype with LangChain, indexing 1,000+ medical docs for cardio/oncology queries; trained an LSTM on 500+ simulated patient timelines to flag early heart and cancer risk indicators.
- Optimized retrieval via ANN search and prompt tuning, achieving sub-2-second average query response time.